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Name - Jai
Rollno. 2301420045

Assignment - 3

Section A

1. What is clocking ?

Clocking methodology is a synchronization technique where all operations within a processor occur based on discrete clock signals. It coordinates the movement of data between registers, memory and functional units by using rising or falling edge of the clock to trigger state changes.

(2) What is clock cycle and why it is important ?

A clock cycle is the smallest unit of time in which a processor operation occurs - one complete oscillation of the clock signal. It is important because it determines the processor's speed. Instructions takes one or multiple cycles to complete, directly impacting performance.

(3) Explain Amdahl's law in computer Architecture.

Amdahl's law states that overall speedup of a system is limited by the non-parallelizable portion of a program. No matter how many processors we add, The performance gain is capped by part that

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cannot be normalized.

Formula - Speedup $= \frac{1}{(1+P) + (P/S)}$

P = parallelization fraction
S = speedup factor on that fraction.

What is data path in a processor?
The data path is a collection of hardware components responsible for processing and moving data within the processor.
It includes the ALU, registers, multiplexers and buses.

What is role of central unit in processor design?
The central unit generates control signals that coordinates operations of ALU, registers and memory.
It acts as traffic cop, decoding instructions and telling each component what to do and when to do.

Explain concept of single-cycle processor.
A single cycle processor executes an entire instruction within one long clock cycle. Its design is simple and easy to control but the clock speed must be enough to accomodate slowest instruction making overall performance poor.

What is multi-cycle processor?
It divides instruction execution into multiple smaller steps (cycles) e.g. Cycle 1: Fetch, Cycle 2: Decode, Cycle 3: Execute etc. Each stage takes one clock cycle, allowing different functional units to be reused and enabling a faster

clock speed and improving efficiency.

What is pipeline in processors?
Pipeline is a process where the execution of multiple instructions is overlapped. The instruction cycle is divided into discrete stages. While one instruction is in the EX stage, the next being decoded and the one after that is being fetched increasing throughput.

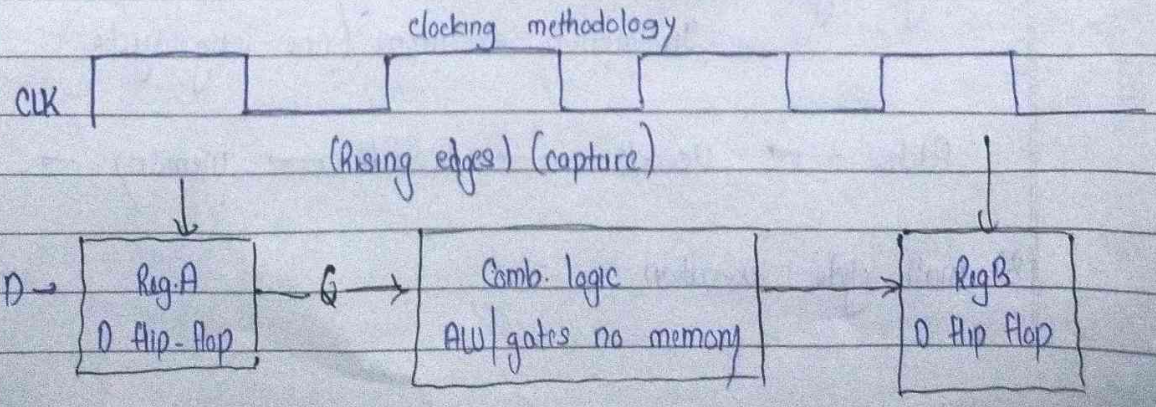
What are control Hazards?
Control hazards occur due to branch instructions that change the program counter (PC). The pipeline does not know which instruction to fetch next until the next branch outcome is completed, potentially leading to wrong instructions being fetched and executed.

(12) Data Hazards – forwarding | bypassing pipeline stalls. compiler instruction reordering

Control Hazards – Branch prediction, delayed branching, branch target buffer, speculative executive with squash.

Section B

1. Basic clocking methodology



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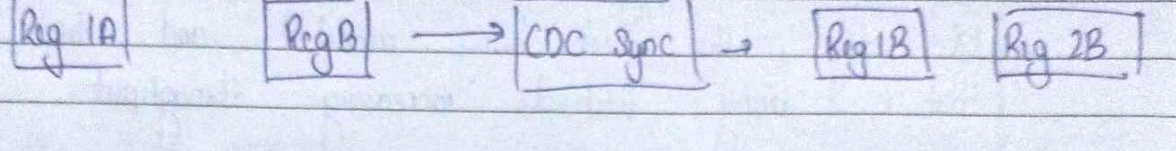
setup time: data stable before rising edge

Singal clock domain — all registers share same CLK signal

Domain A (100 MHz) Domain B (200 MHz)



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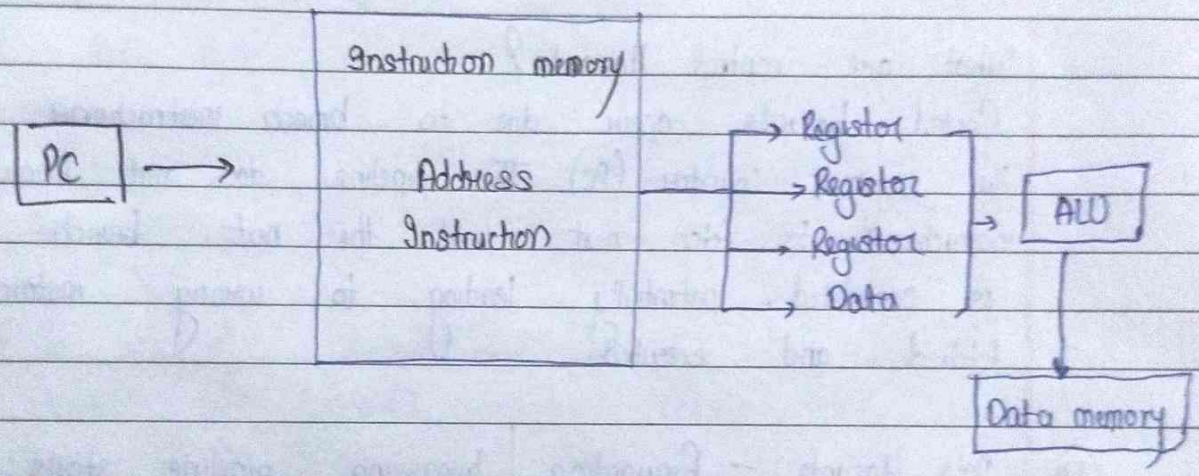

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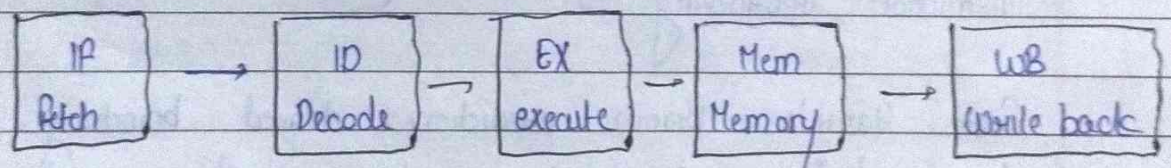

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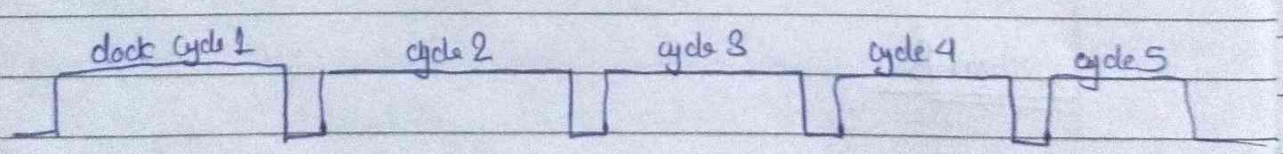

(A) Single Cycle execution

Instruction execution (one long Cycle)

Fetch → Decode → execute → memory → write back

(B) multi cycle execution

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Fetch Instruction → Decode / Reg. Fetch → Execute / Address → Memory Access → Write back

Case Based Analytical Solutions

1. Problem – Data Hazards (RAW – Read after write). The dependent instruction needs a register before the previous instruction has written its result, causing incorrect data.

Solution – Forwarding (Bypassing) – send the ALU result directly to the next instruction input without waiting for write back. Also, compiler scheduling or pipeline stalls if forwarding is not possible.

(2) $P = 0.80$ (parallelizable fraction)
$S = 4$ (speedup on fraction)
sequential fraction $= 1 - P = 0.20$

$$\text{speedup} = \frac{1}{(1-P) + (P/S)} = \frac{1}{(.20) + (.80/4)} \Rightarrow 2.5$$

Overall speedup = 2.5

(3) In single cycle processor, the clock cycle must be long enough for the slowest instruction. All instructions wait for the worst case delay.

In multi-cycle processor, each stage is short and balanced. Fast instructions complete in fewer cycles and the clock run much faster. Hardware resources are reused across cycles improving utilization.

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4. Hazard - Control hazard. The pipeline fetches the next sequential instruction but the branch may change the PC. Incorrect instructions enter the pipeline and must be flushed.

Reduction: Branch prediction, delayed branching, branch target buffer to fetch predicted target early or speculative execution.

Section D:

① Cycle :

	1	2	3	4	5
I1	IF	ID	EX	Mem	WB
I2		IF	ID	EX	Mem
I3			IF	ID	Ex

The EX stage performs the actual computation. For an ALU instruction, the EX stage takes the two register values read in ID, passes them through ALU with the appropriate operation and produces a result

For a load/store instruction, EX calculates the memory address by adding base register and offset.

The result is the forward to Mem stage.

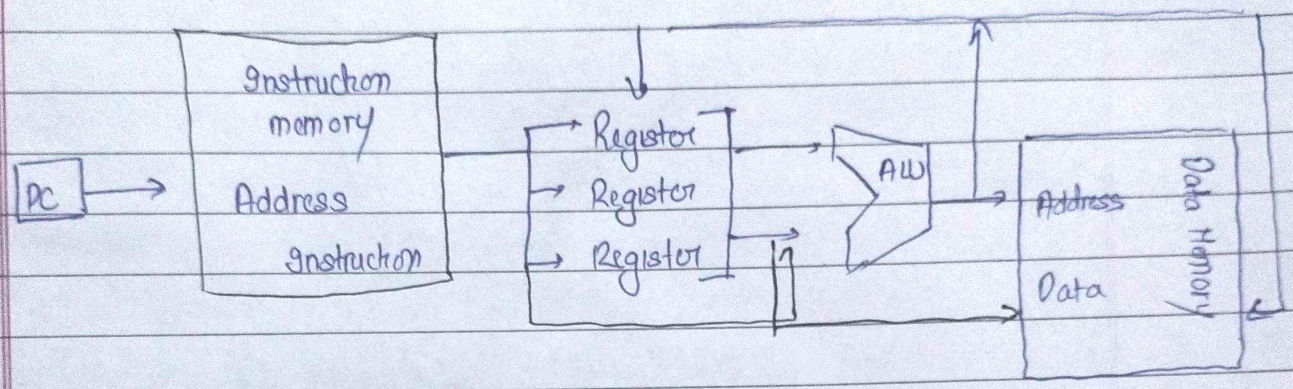
② Data Hazard Scenerio:

add 2, 3, 4
sub 5, 2, 6

Mitigation → Forwarding: The ALU result from Instruction 1

(end of EX stage) is routed directly to input of ALU for instruction 2 (beginning of EX stage) bypassing the register file write back. No stall needed.

③ Datapath of simple CPU and one design decision.



In a pipeline processor, using separate memory allows the IF stage to fetch to next instruction while the MEM stage accesses data simultaneously. This avoids structural hazards and improves throughput. The trade-off is increased hardware complexity.